Pratik Marudwar

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Professional Summary

Data Scientist with 2+ years training, fine-tuning and shipping production ML and LLM systems across healthcare and fintech. Experience spans transformer pre-training and fine-tuning (RoBERTa DAPT, BERT, LoRA/QLoRA), classical ML and statistical inference (propensity score matching, ANCOVA, hierarchical clustering). Comfortable owning a model end to end — dataset curation, training, evaluation, deployment and measurement. M.Tech in Artificial Intelligence, IIT Patna.

Skills

ML & Deep Learning: PyTorch, Transformers, Hugging Face, Scikit-Learn, XGBoost, Gradient Boosting, ANN
LLM Training & Fine-Tuning: LoRA/QLoRA, DAPT, SFT, Quantization, RoBERTa, BERT
Statistics & Experimentation: Propensity Score Matching, ANCOVA, causal inference, pre/post analysis, clustering
Languages & Data: Python, R, SQL, PySpark, Pandas, NumPy, BigQuery, PostgreSQL
Agentic & LLM Frameworks: LangGraph, LangChain, CrewAI, MCP, Anthropic SDK, OpenAI SDK
RAG & Evaluation: ChromaDB, pgvector, Pinecone, FAISS, RAGAS, LangSmith, LLM-as-a-Judge
MLOps & Serving: Django, FastAPI, Docker, Kubernetes, vLLM, Ray Serve, AWS, GitHub Actions, REST APIs

Experience

WebMD — **Dec 2024 – Present**
Associate Data Scientist — *Remote, India*

- Built **Prescriber Lens**, a six-agent **LangGraph** NL2SQL system converting plain-English requests from pharma sales teams into validated **BigQuery** SQL over a healthcare claims cleanroom — intent and entity extraction (drug, condition, ICD-10/J/CPT codes, date ranges), schema selection, **ChromaDB** embedding-based few-shot retrieval, generation, **SQLGlot** AST validation with capped retries, and human-in-the-loop review — cutting manual query effort by **60%** at **85%+ accuracy**.
- Designed campaign ROI measurement using **Propensity Score Matching** to build balanced test/control HCP cohorts, **ANCOVA** for covariate-adjusted lift, and time-aligned pre/post analysis on large-scale prescription (NRx/TRx) data in **BigQuery** and **R**.
- Automated the end-to-end measurement pipeline with scalable Rx and engagement data transformations, cutting analysis turnaround by **60%** (5 days to 2) and removing repeat manual scripting.
- Built an automated geospatial clustering pipeline generating HCP geodomes using **hierarchical distance-based clustering**, replacing per-campaign manual territory analysis.

Rakuten India — **May 2024 – Dec 2024**
Trainee Data Scientist — *Bengaluru, India*

- Performed **domain-adaptive continued pretraining (DAPT)** of **RoBERTa** on Rakuten merchant and financial text corpora in **PyTorch**, then applied supervised fine-tuning for merchant name classification — improving accuracy from **88.05% to 91.35%** over the JLBERT baseline.
- Fine-tuned Japanese LLMs for short-text transaction classification and shipped **production inference APIs adopted by 5 Rakuten business units**.

Projects

Healthcare Professional Detection from Advertisement Data | *Scikit-Learn, XGBoost, Python* — **Jul 2023 – Aug 2023**

- Engineered a multi-model ML pipeline for HCP detection with feature encoding across 13 variables (One-Hot, Target, Frequency, CountVectorizer), achieving **99.7% accuracy** and **0.9965 F1-score** on advertiser interaction data.
- **Models compared:** Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, XGBoost, ANN.

MediAssist AI | *LangGraph, Multi-Agent Orchestration, Django, Python* | GitHub — **May 2026 – Present**

- Built a **nine-agent** LLM platform executing long-horizon, multi-step patient workflows — intake, scheduling, clinical triage, refill coordination, referral execution, prior authorization, outreach, care-gap closure and a 24×7 after-hours orchestrator — across chat, SMS and WhatsApp with FHIR-based EHR integration.
- Designed closed-loop triage agents performing adaptive risk stratification (Emergency/High/Medium/Low/Minimal acuity), with state tracking across multi-turn, multi-day workflows that suspend on human approval and resume hours or days later.

Research

Healthcare Complaint Mining and Classification — **Jul 2023 – Apr 2024**
M.Tech Thesis, Indian Institute of Technology, Patna

- Curated a **10,000-record** annotated dataset via web scraping and structured human annotation across 9 complaint categories, designing the annotation guidelines and label schema.
- Fine-tuned **BERT**-based transformer models in PyTorch for binary and multi-label complaint classification, achieving **82.95% accuracy** on binary complaint detection.

Education

Indian Institute of Technology, Patna — **CGPA: 8.02/10**
Master of Technology, Computer Science (Specialization: Artificial Intelligence) — *Jul 2022 – May 2024*

Maharashtra Institute of Technology, Pune — **CGPA: 8.24/10**
Bachelor of Technology — *Aug 2016 – Nov 2020*